Exploring Brain-Inspired Functional Diversity and Competitive Sparsity in Artificial Neural Networks

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Abstract. While biological nervous systems exhibit rich functional diversity and competitive inhibition, modern artificial neural networks rely predominantly on homogeneous activations and soft gating. Although numerous bio-inspired modifications have been proposed, their practical utility under rigorous, controlled conditions remains poorly understood. In this work, we present a systematic empirical investigation of three parameter-free mechanisms: Multi-Activation Neurons, Neuronal Heterogeneity, and k-Winners-Take-All (kWTA) competitive sparsity. We evaluate MLPs on tabular datasets (Adult Income, Dry Bean) and ConvNeXt V2-Base on image benchmarks (DeepDetect-2025, Tiny ImageNet). On tabular tasks, performance differences are mostly within seed-level variance. On vision tasks, chunked heterogeneity achieves 93.72% accuracy vs. 90.64% baseline on binary classification, while 75%-retention kWTA improves multiclass accuracy to 50.61% vs. 49.81% baseline. However, these mechanisms introduce a 23–134% training runtime overhead and with some configurations showing substantial seed-to-seed variance. Rather than offering universally superior defaults, these mechanisms act as task-dependent inductive biases that require careful, task-specific tuning of structural diversity and sparsity ratios to yield practical utility. Code is available at `https://github.com/rawsie-dev/brain-inspired-neural-networks`.

1 Introduction

Modern deep learning architectures rely on uniform computational primitives, typically applying identical, pointwise non-linear activation functions across a given layer. In contrast, biological neural networks exhibit profound heterogeneity across molecular, morphological, and functional dimensions [19,4]. Dendritic structures perform complex local computations prior to global integration [17,21], while lateral inhibition induces competitive dynamics such as k-Winners-Take-All (kWTA) selection [22,5]. However, whether translating these biologically inspired mechanisms into artificial networks yields tangible computational advantages—or merely introduces unhelpful complexity—remains an open question.

In this work, we conduct a controlled empirical evaluation of three parameter-free abstractions of biological computation: *Multi-Activation Neurons* (intra-neuron non-linear aggregation), *Neuronal Heterogeneity* (inter-neuron functional

diversification), and *k-Winners-Take-All* (activation-level competitive sparsity). Rather than tuning isolated architectures for peak performance, we evaluate these mechanisms under a unified protocol across tabular and vision benchmarks using Multi-Layer Perceptrons (MLPs) and ConvNeXt V2-Base models.

Our primary contributions are fourfold:

- **Rigorous Parameter-Free Formulation:** We isolate the intrinsic computational effect of intra-neuron diversity, inter-neuron diversity, and channel-wise competition without introducing confounding learnable parameters.
- **Multi-Dimensional Empirical Evaluation:** We systematically evaluate these mechanisms across two modalities (tabular vs. vision) and target paradigms (binary vs. multiclass classification).
- **Structural Granularity Analysis:** We assess the role of grouping strategies (random vs. chunked activation assignments), demonstrating that structured inter-neuron heterogeneity preserves signal distinctiveness whereas intra-neuron aggregation induces destructive gradient interference.
- **Realistic Cost-Benefit Analysis:** We demonstrate that parameter-free models incur significant runtime latency (up to 134%), providing a cautionary empirical baseline for future bio-inspired designs.

2 Related Works

Prior work has explored functional diversity in artificial neural networks through learned or evolved activation functions. Choudhary et al. [2] learned neuron-specific activation functions and demonstrated benefits on image classification and nonlinear regression, while Hagg et al. [8] evolved heterogeneous activation functions. Multi-activation approaches have similarly combined multiple nonlinearities within individual neurons through adaptive selection [6] or learned coefficients [9]. In contrast, our mechanisms use fixed, parameter-free assignments to isolate the effect of functional diversity.

Competitive sparsity has been studied through k-Winners-Take-All networks [16], sparse autoencoders [18], and modern architectures [24]. These works motivate explicit competition, while we examine fixed retention levels across tabular and image classification. Neuroscience accounts of lateral inhibition and priority maps [22,5] similarly describe competition that leaves only the most salient responses active, providing the biological basis we abstract as kWTA's fixed retention rule.

Our study therefore differs primarily in its unified comparison: all mechanisms are parameter-free and evaluated under the same experimental framework across two modalities and both binary and multiclass tasks.

3 Background and Motivation

Biological neurons exhibit diversity within cells and across populations, as well as competitive and sparse activity. We abstract these properties as *intra-neuron*

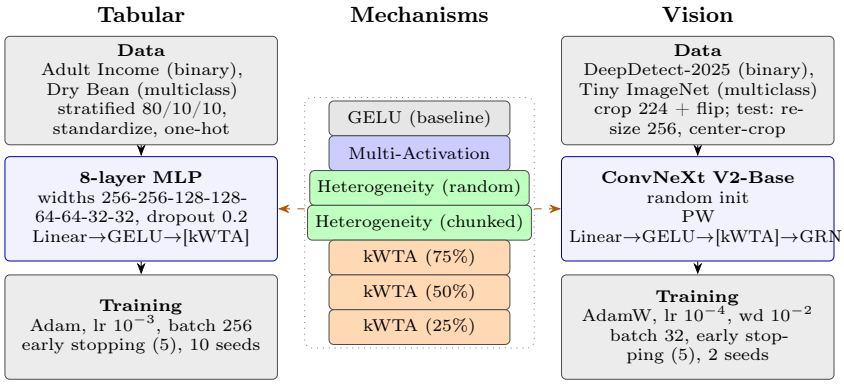


Fig. 1. Experimental pipeline. In each model the GELU is kept (baseline), replaced (Multi-Activation, Heterogeneity), or followed by kWTA (dashed). Variants within each modality share the same data splits, optimizer, and early-stopping protocol.

functional diversity (Multi-Activation), *inter-neuron functional diversity* (Heterogeneity), and *activation-level competition* (kWTA). These are computational analogies rather than biophysically faithful models.

Conventional artificial neurons apply a single nonlinear function, whereas biological neurons exhibit diverse nonlinear response characteristics arising from receptor dynamics, dendritic integration, and voltage-dependent processes [14]. This motivates Multi-Activation, which combines multiple nonlinear transformations within a neuron; the resulting set is intended to span diverse response profiles rather than constitute a theoretically optimal basis (Section 4.1 details the specific functions used). Functional diversity can also occur across neurons in their molecular, morphological, and electrophysiological properties, motivating Heterogeneity, where neurons in the same layer apply different fixed activations [2]. Finally, competitive neural populations exhibit lateral inhibition and selective attention that suppress weaker responses in favor of the strongest [22,5], motivating kWTA, which retains only the strongest activations and discards the rest.

4 Methodology

Figure 1 summarizes the experimental setup.

All three mechanisms are parameter-free: they add no trainable parameters and leave network width and depth unchanged. On vision backbones, kWTA adds the least wall-clock overhead (∼23–28%), followed by Heterogeneity (∼38–40%) and Multi-Activation (∼118–134%).

4.1 Multi-Activation Neurons

The proposed *Multi-Activation Neuron* applies this principle of combining several nonlinear transformations within each neuron.

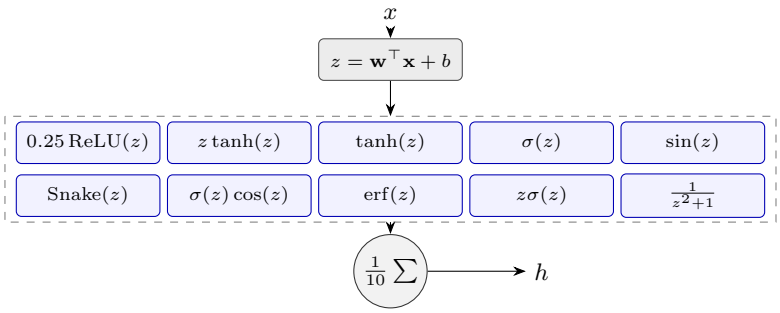


Fig. 2. Multi-Activation Neuron: z is passed through ten fixed activations whose outputs are averaged into h.

The Multi-Activation design is illustrated in Figure 2. For a pre-activation z, the neuron evaluates ten fixed nonlinear functions and averages their outputs.

$$y = \sum_{k=1}^{K} \alpha_k f_k(z). \tag{1}$$

The specific function used in our implementation is:

$$\begin{aligned} f_{\mathrm{MA}}(x) = \frac{1}{10}\big[&0.25\,\mathrm{ReLU}(x) + x\tanh(x) + \tanh(x) + \sigma(x) \\ &+ \sin(x) + G(x) + \mathrm{Snake}(x) + \sigma(x)\cos(x) \\ &+ \mathrm{erf}(x) + x\sigma(x)\big], \end{aligned} \tag{2}$$

where

$$G(x) = \frac{1}{x^2+1}. \tag{3}$$

The ten functions span five qualitative response types: unbounded nondecreasing (ReLU), saturating (tanh, σ, erf), oscillatory (sin, $\sigma(z)\cos(z)$), locally symmetric ($1/(z^2+1)$), and input-dependent modulation ($z\tanh(z)$, Snake(z) [25], $z\sigma(z)$). We scale the unbounded ReLU component by 0.25 to prevent linear drift from dominating the bounded and periodic components at large positive pre-activations. All coefficients are fixed during training.

4.2 Neuronal Heterogeneity

Figure 3 illustrates the two assignment strategies used for Neuronal Heterogeneity.

Conventional neural networks largely omit this cross-neuron diversity by using a shared activation function across each layer [2].

For pre-activations $\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$, each neuron applies one of four activation functions: ReLU, GELU [10], SiLU [7], or Mish [20]. We select these functions to

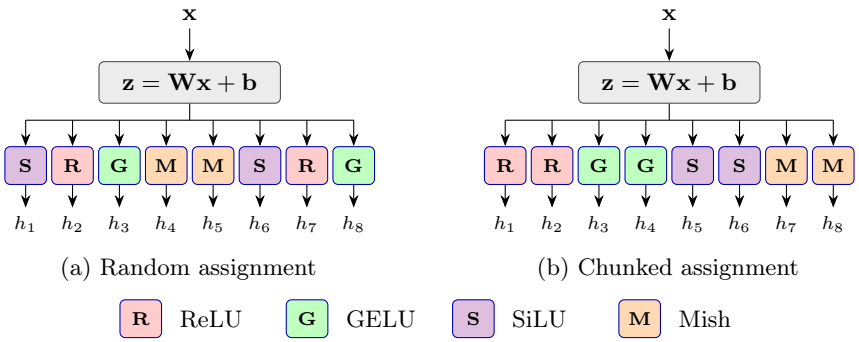


Fig. 3. Neuronal Heterogeneity: each neuron applies a fixed $f_i \in \{\text{ReLU}, \text{GELU}, \text{SiLU}, \text{Mish}\}$, assigned randomly (a) or in contiguous blocks (b) at initialization.

provide a compact set of commonly used nonlinearities with distinct response profiles: ReLU is piecewise-linear and non-saturating, while GELU, SiLU, and Mish provide smooth input-dependent gating through different functional forms. This set is intended to introduce functional diversity without relying on an arbitrarily large collection of activations.

$$f_i \in \{\text{ReLU}, \text{GELU}, \text{SiLU}, \text{Mish}\},$$

giving

$$h_i = f_i(z_i).$$

For the *random* strategy, the four activations are distributed as evenly as possible across neurons and randomly shuffled to avoid systematic grouping. In the *chunked* strategy, neurons are divided into contiguous blocks, with each block assigned a different activation. The assignment is generated once at initialization and remains fixed during training.

4.3 *k*-Winners-Take-All

Figure 4 illustrates the kWTA operation, including the selection of the top $k\%$ of activations and suppression of the remainder.

Unlike the preceding mechanisms, which introduce functional diversity within or across neurons, competitive sparsity controls which activations remain active. Neural representations can exhibit sparse and competitive activity, in which only a subset of neurons responds strongly to a given stimulus. ReLU induces sparsity by setting negative pre-activations to zero, whereas GELU applies smooth, input-dependent gating that attenuates negative and small-magnitude responses. Neither activation explicitly controls the fraction of active neurons.

k-Winners-Take-All (kWTA) introduces explicit competition by retaining only the strongest fraction of activations and suppressing the remainder.

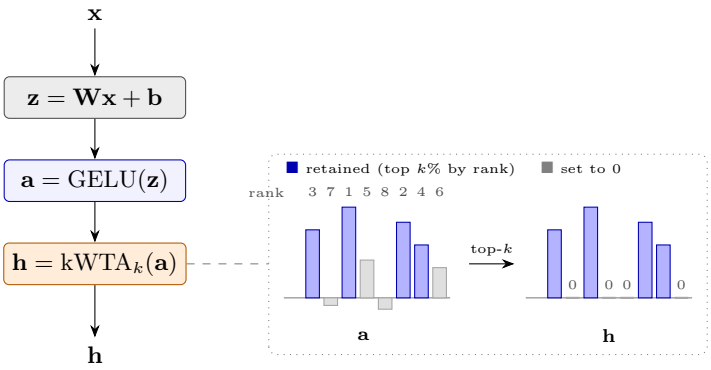


Fig. 4. k-Winners-Take-All after GELU: only the top $k\%$ of activations by value are kept and the rest (including small negative GELU outputs) are set to 0 (shown for $k = 50\%$). In ConvNeXt V2 it sits between GELU and GRN.

$$\mathrm{kWTA}_k(\mathbf{a})_i = \begin{cases} a_i, & a_i \text{ is among the largest } k\% \text{ of entries of } \mathbf{a}, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Unlike the two activation-based mechanisms, kWTA does not change the nonlinear transformation of individual neurons. Instead, it controls interactions between activations by imposing a specified level of sparsity. Retained activations preserve their magnitudes without renormalization.

During backpropagation, gradients are propagated exclusively through the winning units and zeroed out for suppressed units:

$$\frac{\partial h_i}{\partial a_i} = \begin{cases} 1, & \text{if } i \in \text{top-}k\%, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Consequently, gradients flow back only through the active pathway determined during the forward pass. The selected k value remains fixed throughout each experiment. Multiple sparsity levels are evaluated to assess the effect of competition strength, retaining the top 75%, 50%, and 25% of activations ($k = 0.75, 0.5, 0.25$).

For both tabular and vision models, kWTA is applied along the feature dimension. Given an activation vector $a \in \mathbb{R}^d$, the operator retains the largest $k\%$ of activations by value and sets the remainder to zero, without renormalization. For ConvNeXt V2, the activation tensor is represented as $X \in \mathbb{R}^{B\times H\times W\times C}$, and kWTA is applied independently to $X_{b,h,w,:} \in \mathbb{R}^C$ at each sample and spatial location. Thus, competition occurs among channels at each spatial position rather than across spatial positions or samples.

5 Experimental Setup

5.1 Datasets

We evaluate four public datasets covering tabular and image classification. Adult Income [1] contains 48,842 instances for binary classification, while Dry Bean [12] contains 13,611 instances across seven classes. DeepDetect-2025 [3] contains 112,185 images for binary classification of real and AI-generated images. Tiny ImageNet [13] contains 100,000 training images across 200 classes for multiclass classification.

5.2 Models and Training

For tabular data, we use an 8-layer MLP with widths 256–256–128–128–64–64–32–32, dropout $p = 0.2$, and a task-specific linear output layer. Adult Income contains 105 preprocessed features. Both datasets use stratified 80/10/10 train/validation/test splits, with numerical features standardized using training statistics and Adult categorical features one-hot encoded. Models are trained from scratch with Adam [11], learning rate 10^{-3} and batch size 256, for up to 100 epochs with validation-loss early stopping (patience 5). Ten seeds are used.

For image data, we use randomly initialized ConvNeXt V2-Base [23]. Within the ConvNeXt V2 block, the proposed mechanisms modify the pointwise nonlinear sub-block: Multi-Activation and Heterogeneity replace GELU, whereas kWTA is inserted after GELU and before GRN.

$$\text{Linear} \rightarrow \text{GELU} \rightarrow \text{GRN} \rightarrow \text{Linear},$$

and the proposed mechanisms modify the GELU operation, except kWTA, which is inserted after GELU. DeepDetect-2025 uses an 81,369/9,040 train/validation split and a 21,776-image test set. Tiny ImageNet is split into 90,000/10,000 training/validation images, with its original validation set used for testing. Training uses random resized cropping to 224×224 and horizontal flipping; testing uses resize-to-256 and center cropping. AdamW [15] is used with learning rate 10^{-4}, weight decay 10^{-2}, batch size 32, and patience-5 early stopping. Two seeds (42, 123) are used.

Binary tasks report loss, accuracy, positive-class F1, and ROC-AUC; multiclass tasks report loss, accuracy, macro-F1, and macro one-vs-rest ROC-AUC.

6 Results and Interpretations

Results are mean ± standard deviation over 10 tabular and 2 vision seeds; large deviations warrant cautious interpretation. For *kWTA*, percentages denote the proportion of activations retained after the activation step; all remaining activations are set to zero.

Table 1. Results on the tabular dataset for binary classification.

Experiment	Test Loss	Test Accuracy	F1	AUC
Baseline	0.3094 ± 0.0061	0.8568 ± 0.0046	0.6705 ± 0.0140	0.9138 ± 0.0027
Multi-Activation Neurons	0.3074 ± 0.0061	0.8565 ± 0.0041	0.6746 ± 0.0107	0.9137 ± 0.0029
Heterogeneity (random)	0.3073 ± 0.0056	0.8588 ± 0.0037	0.6771 ± 0.0143	0.9152 ± 0.0026
Heterogeneity (chunked)	0.3070 ± 0.0059	0.8574 ± 0.0037	0.6742 ± 0.0132	0.9149 ± 0.0027
kWTA (75%)	0.3112 ± 0.0062	0.8571 ± 0.0041	0.6774 ± 0.0125	0.9137 ± 0.0026
kWTA (50%)	0.3111 ± 0.0066	0.8557 ± 0.0046	0.6521 ± 0.0176	0.9133 ± 0.0027
kWTA (25%)	0.3117 ± 0.0050	0.8554 ± 0.0051	0.6552 ± 0.0117	0.9127 ± 0.0024

Table 2. Results on the tabular dataset for multiclass classification.

Experiment	Test Loss	Test Accuracy	Macro F1	Macro AUC
Baseline	0.2144 ± 0.0264	0.9280 ± 0.0075	0.9388 ± 0.0077	0.9946 ± 0.0013
Multi-Activation Neurons	0.2281 ± 0.0222	0.9256 ± 0.0059	0.9370 ± 0.0049	0.9938 ± 0.0010
Heterogeneity (random)	0.2096 ± 0.0213	0.9296 ± 0.0060	0.9404 ± 0.0060	0.9948 ± 0.0010
Heterogeneity (chunked)	0.2163 ± 0.0224	0.9277 ± 0.0061	0.9385 ± 0.0059	0.9945 ± 0.0011
kWTA (75%)	0.2210 ± 0.0269	0.9275 ± 0.0080	0.9384 ± 0.0079	0.9943 ± 0.0013
kWTA (50%)	0.2188 ± 0.0272	0.9275 ± 0.0073	0.9388 ± 0.0078	0.9946 ± 0.0012
kWTA (25%)	0.2616 ± 0.0358	0.9195 ± 0.0090	0.9337 ± 0.0079	0.9931 ± 0.0014

6.1 Multi-Activation Neurons

Multi-Activation did not consistently improve upon the baseline. Tabular performance shifts remained within one standard deviation. On binary image classification, accuracy and F1 increased to 91.81% and 0.9169 (baseline: 90.64% and 0.9037). Conversely, on multiclass image classification, accuracy and macro F1 dropped to 46.78% ± 0.66% and 0.4628 (baseline: 49.81% and 0.4955). This limitation may stem from feature cancellation during intra-neuron aggregation, where summing non-linearities within a single neuron smooths feature boundaries and hampers optimization.

6.2 Neuronal Heterogeneity

Neuronal Heterogeneity showed strong performance across tasks. For tabular binary classification, random heterogeneity led in accuracy (85.88%) and AUC (0.9152), chunked heterogeneity achieved the lowest loss (0.3070), and kWTA (75%) recorded the top F1 (0.6774). On tabular multiclass, random heterogeneity yielded the best metrics across all evaluations (92.96% accuracy, 0.2096 loss).

The largest gain occurred in binary image classification, where chunked heterogeneity achieved peak performance (93.72% accuracy, 0.1949 loss vs. 90.64% accuracy, 0.2718 loss for baseline), indicating that assigning distinct activations per neuron enables sub-path specialization. On multiclass image classification, chunked heterogeneity improved over the baseline (50.47% accuracy), though kWTA (75%) achieved the top overall result (50.61% accuracy, 2.0903 loss).

Table 3. Results on the image dataset for binary classification.

Experiment	Test Loss	Test Accuracy	F1	AUC
Baseline	0.2718 ± 0.0294	0.9064 ± 0.0108	0.9037 ± 0.0123	0.9840 ± 0.0025
Multi-Activation Neurons	0.2273 ± 0.0401	0.9181 ± 0.0145	0.9169 ± 0.0149	0.9850 ± 0.0080
Heterogeneity (random)	0.2295 ± 0.0447	0.9271 ± 0.0249	0.9267 ± 0.0260	0.9821 ± 0.0041
Heterogeneity (chunked)	0.1949 ± 0.0020	0.9372 ± 0.0184	0.9376 ± 0.0187	0.9857 ± 0.0038
kWTA (75%)	0.3596 ± 0.0236	0.8941 ± 0.0141	0.8899 ± 0.0153	0.9773 ± 0.0053
kWTA (50%)	0.3277 ± 0.1853	0.9085 ± 0.0482	0.9049 ± 0.0537	0.9812 ± 0.0099
kWTA (25%)	0.2878 ± 0.0846	0.9115 ± 0.0139	0.9099 ± 0.0147	0.9791 ± 0.0120

Table 4. Results on the image dataset for multiclass classification.

Experiment	Test Loss	Test Accuracy	Macro F1	Macro AUC
Baseline	2.1515 ± 0.0127	0.4981 ± 0.0061	0.4955 ± 0.0058	0.9717 ± 0.0003
Multi-Activation Neurons	2.2896 ± 0.0055	0.4678 ± 0.0066	0.4628 ± 0.0049	0.9683 ± 0.0007
Heterogeneity (random)	2.1450 ± 0.0025	0.4975 ± 0.0012	0.4940 ± 0.0037	0.9718 ± 0.0001
Heterogeneity (chunked)	2.1338 ± 0.0093	0.5047 ± 0.0057	0.5004 ± 0.0038	0.9725 ± 0.0007
kWTA (75%)	2.0903 ± 0.0139	0.5061 ± 0.0090	0.5027 ± 0.0105	0.9742 ± 0.0004
kWTA (50%)	2.1147 ± 0.0219	0.5033 ± 0.0056	0.4992 ± 0.0062	0.9733 ± 0.0006
kWTA (25%)	2.1186 ± 0.0066	0.5010 ± 0.0112	0.4974 ± 0.0147	0.9735 ± 0.0001

6.3 *k*-Winners-Take-All

kWTA showed a retention-dependent effect on tabular data (25% was lowest) that was not resolvable on vision. On tabular binary classification, 75% retention gave the best kWTA accuracy (85.71%) and the highest F1 in the entire binary tabular comparison (67.74%), while 25% gave the worst accuracy (85.54%) and 50% gave the worst F1 (65.21%). On tabular multiclass classification, 50% retention preserved the baseline macro F1 (93.88%) and macro AUC (99.46%), whereas 25% caused a clearer drop in accuracy and macro F1.

On binary image classification, 25% retention reached 91.15% accuracy and 50% reached 90.85% accuracy and 90.49% F1, but the 50% setting varied strongly across runs (standard deviations of 4.82 and 5.37 percentage points). On multiclass image classification, all three kWTA settings exceeded the baseline accuracy, with 75% retention best (50.61% accuracy, 50.27% macro F1).

One possible explanation for this contrast is the mechanism's gradient dynamics. Because suppressed units receive zero gradients ($\partial h_i/\partial a_i = 0$), stacking kWTA across an 8-layer MLP removes 75% of gradient paths at each layer when using 25% retention. This may make optimization more difficult in earlier layers. In ConvNeXt V2, residual connections provide additional shortcut pathways around local channel-wise suppression, which may reduce this effect on vision tasks.

6.4 Computational Overhead and Practical Efficiency

While all mechanisms are parameter-free, their execution costs differ substantially on ConvNeXt V2-Base (Table 5). Multi-Activation increased epoch time by approximately 118–134%, reflecting the evaluation of ten nonlinearities, while

Table 5. Empirical training duration per epoch across tasks and mechanisms. Tabular models were evaluated on a multi-core CPU, while image models were trained on an NVIDIA GeForce RTX 5090 GPU.

Mechanism	Tabular Binary (sec/epoch)	Tabular Multiclass (sec/epoch)	Image Binary (min/epoch)	Image Multiclass (min/epoch)
Baseline	< 1 s	< 1 s	7m 54s	8m 41s
Multi-Activation Neurons	< 1 s	< 1 s	18m 28s	18m 56s
Heterogeneity (random)	< 1 s	< 1 s	10m 58s	12m 09s
Heterogeneity (chunked)	< 1 s	< 1 s	10m 53s	12m 04s
kWTA (75%)	< 1 s	< 1 s	09m 55s	11m 05s
kWTA (50%)	< 1 s	< 1 s	09m 51s	10m 54s
kWTA (25%)	< 1 s	< 1 s	09m 43s	10m 50s

Heterogeneity and kWTA increased it by approximately 38–40% and 23–28%, respectively. Tabular runtimes remained below one second per epoch for all variants. Thus, parameter count alone does not capture the computational cost of these mechanisms.

6.5 Differences Across Modalities

The mechanisms generally produced smaller performance differences on tabular data and larger differences on image tasks. One possible explanation is that tabular features are compact and explicitly defined, making activation suppression more likely to remove useful information, whereas hierarchical image representations may provide more opportunity for functional specialization and competitive selection.

This pattern is particularly visible for kWTA: 25% retention was the weakest setting on both tabular datasets, while all three retention levels matched or exceeded the baseline on image multiclass classification, with 75% performing best. On image binary classification, kWTA did not clearly improve performance, with differences largely within seed variability.

The multiclass image task remained difficult, with models achieving approximately 46–51% accuracy despite macro-AUC values near 0.97. This illustrates that AUC and top-1 accuracy capture different properties, with AUC showing smaller between-seed variation in these experiments.

6.6 Overall Interpretation

Across the four evaluation settings, no mechanism was universally superior. Multi-Activation showed no improvement on tabular data (small nominal decreases within one standard deviation), a nominal gain on image binary, and lower accuracy on image multiclass. This does not rule out intra-neuron diversity as a mechanism, since alternative activation sets, combination rules, or learned weightings remain unexplored; the results characterize the specific aggregation strategy evaluated rather than the underlying hypothesis.

Neuronal Heterogeneity showed the most consistently competitive results, particularly chunked heterogeneity on binary image classification. kWTA produced modest gains on image multiclass classification, with 75% retention reaching 50.61% accuracy versus 49.81% for the baseline, although this difference is approximately one standard deviation and is based on only two seeds. On image binary classification, kWTA did not clearly improve performance, while aggressive 25% sparsity was harmful on tabular data. Overall, the observed effects depend on both task and mechanism configuration, and seed variance prevents distinguishing many small effects from noise.

7 Discussion and Future Directions

7.1 Diversity Within Versus Across Neurons and Its Implications

Multi-Activation and Heterogeneity distribute functional diversity differently: the former combines nonlinearities within each neuron, whereas the latter assigns one nonlinearity per neuron. The stronger results of Heterogeneity in several settings are consistent with the possibility that distributing nonlinear responses preserves more distinct response patterns. However, the experiments do not establish this mechanism, and alternative activation sets or aggregation rules may produce different outcomes.

A natural extension is a parameterized mixture:

$$y_i = \sum_{k=1}^{K} \alpha_{ik} f_k(z_i), \tag{6}$$

where α_{ik} controls the contribution of each activation. This framework could interpolate between distributed and aggregated functional diversity while allowing the contributions to be learned.

7.2 Competitive Sparsity and Sparsity Level

kWTA provides a different form of biological inspiration by controlling which activations propagate rather than changing their nonlinearities. Its results show that sparsity should be treated as a tunable inductive bias rather than an objective to maximize.

As noted in Section 6.5, the best retention level differed between tasks (75% on Tiny ImageNet, 25% among kWTA settings on DeepDetect-2025). Notably, the 50% binary-image runs differed by 6.81 percentage points between seeds, demonstrating that stochasticity can be substantial. Consequently, individual best-case results cannot establish the superiority of a particular sparsity ratio without repeated trials. Future experiments should first evaluate promising configurations across multiple seeds and additional datasets to establish whether the observed improvements are reproducible.

7.3 Limitations

Several limitations should be considered when interpreting these results. First, the proposed mechanisms introduce computational overhead. In particular, the absence of specialized fused CUDA kernels for the heterogeneous and Multi-Activation operations may require multiple separate memory operations, increasing training latency. This overhead is an implementation limitation rather than an inherent requirement of the mechanisms and could potentially be reduced through kernel fusion.

Second, the evaluation scale is limited. Due to the substantial computational cost of training the 89M-parameter backbone from scratch across multiple epochs and mechanisms, vision experiments use only two fixed seeds (42, 123). These runs therefore characterize initialization sensitivity and structural trends rather than providing definitive statistical evidence. The study also does not evaluate Vision Transformers or other architectural families. Broader evaluation across architectures and more seeds would provide stronger evidence of generality.

Finally, the activation assignments and kWTA retention ratios are fixed using predefined heuristics rather than learned automatically. While this enables controlled comparisons between mechanisms, learned activation selection or gating could potentially adapt these choices to different layers or tasks. Future work should therefore investigate learnable formulations alongside broader architectural and computational evaluations.

8 Conclusion

This empirical reality check demonstrates that biological plausibility alone does not grant deep learning gains. Rather, practical utility depends on functional structure, data modality, and execution overhead:

- **Structure Matters:** Distributing nonlinearities across neurons (inter-neuron heterogeneity) produced more favorable results than the evaluated intra-neuron aggregation strategy, particularly on Tiny ImageNet, where Heterogeneity (chunked) reached 50.47% accuracy compared with 46.78% for Multi-Activation.
- **Modality-Dependent Sparsity:** The effect of kWTA depended on both modality and retention level. Aggressive 25% retention was detrimental on tabular data, while 75% retention produced a modest gain on Tiny ImageNet relative to the baseline.
- **Parameter-Free Does Not Mean Cost-Free:** Although these mechanisms add no trainable parameters, they increased ConvNeXt V2-Base training time by approximately 23%–134%, illustrating that parameter count alone does not capture their computational cost.

Ultimately, neuro-inspired design requires structured diversity and modality-aware sparsity balanced against actual hardware efficiency.

References

1. Becker, B., Kohavi, R.: Adult. UCI Machine Learning Repository (1996). https://doi.org/10.24432/C5XW20
2. Choudhary, A., Radhakrishnan, A., Lindner, J.F., Sinha, S., Ditto, W.L.: Neuronal diversity can improve machine learning for physics and beyond. Scientific Reports **13**, 13962 (2023). https://doi.org/10.1038/s41598-023-40766-6
3. Datta, A.: DeepDetect-2025 (2025). https://doi.org/10.34740/KAGGLE/DSV/11658923, https://www.kaggle.com/dsv/11658923
4. Di Bella, D.J., Domínguez-Iturza, N., Brown, J.R., Arlotta, P.: Making Ramón y Cajal proud: Development of cell identity and diversity in the cerebral cortex. Neuron **112**(13), 2091–2111 (2024). https://doi.org/10.1016/j.neuron.2024.04.021
5. Duncan, D.H., van Moorselaar, D., Theeuwes, J.: Pinging the brain to reveal the hidden attentional priority map using encephalography. Nature Communications **14**(1), 4749 (2023). https://doi.org/10.1038/s41467-023-40405-8
6. Dushkoff, M., Ptucha, R.: Adaptive activation functions for deep networks. In: Electronic Imaging. vol. 28, pp. 1–5 (2016). https://doi.org/10.2352/ISSN.2470-1173.2016.19.COIMG-149
7. Elfwing, S., Uchibe, E., Doya, K.: Sigmoid-weighted linear units for neural network function approximation in reinforcement learning. Neural Networks **107**, 3–11 (2018). https://doi.org/10.1016/j.neunet.2017.12.012
8. Hagg, A., Mensing, M., Asteroth, A.: Evolving parsimonious networks by mixing activation functions. arXiv preprint arXiv:1703.07122 (2017)
9. Harmon, M., Klabjan, D.: Activation ensembles for deep neural networks. arXiv preprint arXiv:1702.07790 (1702)
10. Hendrycks, D., Gimpel, K.: Gaussian error linear units (GELUs). In: International Conference on Learning Representations (ICLR) (2018)
11. Kingma, D.P., Ba, J.: Adam: A method for stochastic optimization. In: International Conference on Learning Representations (ICLR) (2015)
12. Koklu, M., Ozkan, I.A.: Multiclass classification of dry beans using computer vision and machine learning techniques. Computers and Electronics in Agriculture **174**, 105507 (2020). https://doi.org/10.1016/j.compag.2020.105507
13. Le, Y., Yang, X.: Tiny ImageNet Visual Recognition Challenge. CS231N course project, Stanford University (2015)
14. London, M., Häusser, M.: Dendritic computation. Annual Review of Neuroscience **28**, 503–532 (2005). https://doi.org/10.1146/annurev.neuro.28.061604.135703
15. Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. In: International Conference on Learning Representations (ICLR) (2019)
16. Majani, E., Erlanson, R., Abu-Mostafa, Y.S.: On the K-winners-take-all network. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 1, pp. 634–642 (1988)
17. Makarov, R., Pagkalos, M., Poirazi, P.: Dendrites and efficiency: Optimizing performance and resource utilization. Current Opinion in Neurobiology **83**, 102812 (2023). https://doi.org/10.1016/j.conb.2023.102812
18. Makhzani, A., Frey, B.: Winner-take-all autoencoders. In: Proceedings of the 32nd International Conference on Machine Learning (ICML). Proceedings of Machine Learning Research, vol. 37, pp. 277–285 (2015)
19. Mao, X., Staiger, J.F.: Multimodal cortical neuronal cell type classification. Pflügers Archiv - European Journal of Physiology **476**, 721–733 (2024). https://doi.org/10.1007/s00424-024-02923-2

20. Misra, D.: Mish: A self regularized non-monotonic activation function. In: British Machine Vision Conference (BMVC) (2020)
21. Pagkalos, M., Chavlis, S., Poirazi, P.: Introducing the Dendrify framework for incorporating dendrites to spiking neural networks. Nature Communications **14**, 131 (2023). https://doi.org/10.1038/s41467-022-35747-8
22. Roach, J.P., Churchland, A.K., Engel, T.A.: Choice selective inhibition drives stability and competition in decision circuits. Nature Communications **14**, 147 (2023). https://doi.org/10.1038/s41467-023-35822-8
23. Woo, S., Debnath, S., Hu, R., Chen, X., Liu, Z., Kweon, I.S., Xie, S.: ConvNeXt V2: Co-designing and scaling ConvNets with masked autoencoders. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 16133–16142 (2023). https://doi.org/10.1109/CVPR52729.2023.01548
24. Xiao, C., Zhong, P., Zheng, C.: Enhancing adversarial defense by k-winners-take-all. In: International Conference on Learning Representations (ICLR) (2020)
25. Ziyin, L., Hartwig, T., Ueda, M.: Neural networks fail to learn periodic functions and how to fix it. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 33, pp. 1583–1594 (2020)